

Multiple Choice (10 marks)

1. In the diamond structure of elemental carbon, the nearest neighbors of each C atom lie at the corners of a
 - (a) square
 - (b) hexagon
 - (c) cube
 - (d) tetrahedron
2. A meter stick with a speed of $0.8c$ moves past an observer. In the observer's reference frame, how long does it take the stick to pass the observer ?
 - (a) 1.6 ns
 - (b) 2.5 ns
 - (c) 4.2 ns
 - (d) 6.9 ns
3. For which of the following thermodynamic processes is the increase in the internal energy of an ideal gas equal to the heat added to the gas?
 - (a) Constant temperature
 - (b) Constant volume
 - (c) Constant pressure
 - (d) Adiabatic
4. The speed of light inside of a nonmagnetic dielectric material with a dielectric constant of 4.0 is
 - (a) 1.2×10^9 m/s
 - (b) 3.0×10^8 m/s
 - (c) 1.5×10^8 m/s
 - (d) 1.0×10^8 m/s
5. A rod measures 1.00 m in its rest system. How fast must an observer move parallel to the rod to measure its length to be 0.80 m?
 - (a) $0.50c$
 - (b) $0.60c$
 - (c) $0.70c$
 - (d) $0.80c$

True / False (10 marks)

Answer True or False

6. True or False: The speed of light in a nonmagnetic dielectric material with a dielectric constant of 4.0 is 1.2×10^9 m/s.
7. True or False: A photon detector with a quantum efficiency of 0.1 will detect exactly 10 photons when 100 photons are sent in.
8. True or False: During a reversible thermodynamic process, the internal energy of the system does not change at all.
9. True or False: The primary source of the Sun's energy is the fusion of four hydrogen atoms into one helium atom, releasing energy according to $E=mc^2$.
10. True or False: In the lab frame, a particle moving at $v=0.60$ c will travel 450 m before it decays.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the harmonic series of an organ pipe that is closed at one end and open at the other. Given that its fundamental frequency is C (131 Hz), calculate and explain the frequency of the next higher harmonic.
12. A police car is moving towards a wall at 3.5 m/s while emitting a siren at 600 Hz. Calculate the frequency of the echo heard by the driver, explaining the Doppler effect involved in your analysis.

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